



Activities & Assessments

Programming and Artificial Intelligence

CHAPTER 3 | PART ONE | FIRST SEMESTER

Unit 3: Web Applications

3 lessons | Performance tasks | Home assignments | Weekly assessments

CHAPTER 3

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Unit 3: Web Applications

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HOW THIS WORKBOOK IS ORGANISED

1

Classroom Performances *(continuous formative assessment)*

Varied activities the teacher uses during the lesson to boost student interaction, check progress directly, and identify strengths and areas needing improvement.

2

Home Assignments *(mastery and self-application)*

Guided activities and exercises that help students learn independently, check their understanding, and apply what they have learned in new contexts.

3

Weekly Assessments *(performance measurement and follow-up)*

Assessment models carried out at the end of each week to track progress and measure the learning outcomes achieved during that week.

Part I of each period contains the classroom *Performance Tasks*; **Part II** contains the *Home Assignments*. Period 3 of every lesson holds the *Weekly Assessments* in three models (A, B and C). Each question has dotted lines for your written answer.

UNIT 3 | WEB APPLICATIONS

LESSON

1

The Overall Structure of Web Applications

PERIOD 1

Week:

Date:/...../.....

Part I | Performance Tasks

Essay Questions:

1. Define a web application.

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2. List the main tiers that make up a web application.

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Multiple-Choice Questions: (choose the correct answer)

1. A web application is usually built from:

A. Three tiers.

C. One tier.

B. Two tiers.

D. Several tiers.

2. The tier responsible for displaying screens and collecting the user's data and actions is:

A. The backend.

C. The frontend.

B. The database.

D. The processing server.

Part II | Home Assignments

Essay Questions:

1. Compare the role of the frontend with the role of the backend in a social-media application.

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2. How does the backend handle data coming from the frontend and data stored in the database, in an e-commerce website?

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Multiple-Choice Questions: (choose the correct answer)

- On a video-sharing site, aggregating the view count and choosing the recommended videos is the work of:
 - The frontend.
 - The backend.
 - The database.
 - The communication network.
- Storing post data, follow relationships, and messages in a social-media network is the function of:
 - The processing server.
 - The frontend.
 - The database.
 - The backend.

PERIOD 2

Week:

Date:/...../.....

Part I | Performance Tasks**Essay Questions:**

- Design a structural diagram showing how the three tiers cooperate to carry out a user's action.

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- Discuss the importance of the database tier on e-commerce websites.

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Multiple-Choice Questions: (choose the correct answer)

- Sending payment requests and handling their results on an e-commerce site is joint work between processing and control in:
 - The frontend.
 - The database.
 - The backend.
 - The user's browser.
- The main role of the database tier in web applications is:
 - Receiving user actions.
 - Storing and retrieving data.
 - Performing calculations.
 - Displaying screens and interfaces.

Part II | Home Assignments**Essay Questions:**

- Explain the overall structure of web applications and the three tiers.

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- Explain how the three tiers integrate to provide a "video-sharing" service, giving an example of the function of each tier.

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Multiple-Choice Questions: (choose the correct answer)

- On a video-sharing site, the record of views and user information is stored in:

A. The search screens.

B. The frontend.

C. The backend.

D. The database tier.
- Controlling instant notifications and identifying the target audience for posts is done through:

A. The backend.

B. The frontend.

C. The database.

D. The timeline.

PERIOD 3**Weekly Assessments**

Week:

Date:/...../.....

MODEL A**Essay Questions:**

- Explain the difference between the frontend and the backend in web applications, giving an example of each.
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- Explain the role of the database in a web application, and how it cooperates with the backend to show results to the user.
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- Trace the steps of a product-search request flowing through the three tiers in an e-commerce store, showing the role of each tier.
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.....
- Design a suitable distribution of responsibilities for an e-commerce web application, and justify your decisions.
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Multiple-Choice Questions: (choose the correct answer)

- Which of the following represents the role of the frontend in a web application?

A. Processing requests.

B. Displaying content and receiving user input.

C. Storing data.

D. Verifying login.
- When searching for a product in an online shop, which tier sends the request to the database?

A. The frontend.

B. The backend.

C. The database itself.

D. The browser.
- Which of the following is an example of data stored in an e-commerce website's database?

A. The search screen.

B. The order history.

C. Login verification.

D. The user interface.

4. The main idea of the overall structure of web applications is:

- A. Building the application from a single tier only.
- B. Three tiers cooperating: frontend, backend, database.
- C. Using only the browser.
- D. Relying on the backend without a database.

MODEL B

Essay Questions:

1. Explain the concept of a web application, giving examples from your daily life.

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2. Discuss the importance of dividing a web application into three main tiers.

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3. Show how the three tiers cooperate when using a social-media network.

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4. Analyze the role of the backend on an e-commerce website.

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Multiple-Choice Questions: (choose the correct answer)

1. Which of the following is an example of a web application?

- A. A video player program on the device.
- B. A maps website on the Internet.
- C. A desktop calculator application.
- D. A photo-editing program.

2. On a video-sharing site, which tier stores video and user data?

- A. The frontend.
- B. The database.
- C. The backend.
- D. The browser.

3. Which of the following represents the function of the backend in a web application?

- A. Displaying product pages.
- B. Displaying the search box for the user.
- C. Verifying login.
- D. Storing the order history.

4. The main idea of the overall structure of web applications is:

- A. The application is built from three tiers that cooperate together.
- B. Each tier works alone without cooperation.
- C. The application works only through the browser.
- D. The application relies only on the database.

MODEL C**Essay Questions:**

1. What is the role of the frontend on a video website?

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2. What is the role of the database in a social-media network?

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3. Identify the steps of cooperation when sending an email.

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4. What is the importance of having a strong backend tier in e-commerce?

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Multiple-Choice Questions: (choose the correct answer)

1. The main idea of the overall structure of web applications is:

- | | |
|---|--|
| A. Each tier works alone without cooperation. | B. The application relies only on the database. |
| C. The application works only through the browser. | D. The application is built from three tiers that cooperate together. |

2. On a video-sharing site, which tier stores video and user data?

- | | |
|-------------------------|------------------------|
| A. The frontend. | B. The backend. |
| C. The database. | D. The browser. |

3. Which of the following is an example of a web application?

- | | |
|---|---|
| A. A video player program on the device. | B. A maps website on the Internet. |
| C. A desktop calculator application. | D. A photo-editing program. |

4. Which of the following represents the function of the backend in a web application?

- | | |
|--------------------------------------|---|
| A. Displaying product pages. | B. Displaying the search box for the user. |
| C. Storing the order history. | D. Verifying login. |

UNIT 3 | WEB APPLICATIONS

LESSON

2

Web Application Communication Methods

PERIOD 1

Week:

Date:/...../.....

Part I | Performance Tasks

Essay Questions:

1. Explain the client-server model.

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2. Design a diagram showing how the client and the server communicate.

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Multiple-Choice Questions: (choose the correct answer)

1. The client in web applications is:

A. The server that processes requests.

C. The database.

B. The program or device that sends the requests.

D. The communication protocol.

2. When sending login data, it is preferable to use:

A. GET

C. 200

B. POST

D. 404

Part II | Home Assignments

Essay Questions:

1. Why is the HTTPS protocol considered more secure than HTTP? Give an example.

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2. Explain the difference between a GET request and a POST request, with an example of each.

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Multiple-Choice Questions: (choose the correct answer)

- Status code 200 means:
 - An internal server error.
 - The request succeeded.
 - The resource does not exist.
 - An unauthorized request.
- Status code 404 indicates:
 - The request succeeded.
 - An internal server error.
 - The resource does not exist.
 - The connection is secure.

PERIOD 2

Week:

Date:/...../.....

Part I | Performance Tasks**Essay Questions:**

- Explain the meaning of status code 200 in HTTP.
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- What does status code 404 mean, and when does it appear?
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Multiple-Choice Questions: (choose the correct answer)

- The basic difference between HTTP and HTTPS is:
 - HTTPS uses TLS encryption.
 - HTTP is always faster.
 - HTTPS does not support POST.
 - HTTP uses JSON.
- The JSON format is used for:
 - Designing the user interface.
 - Exchanging structured data.
 - Encrypting the connection.
 - Setting the status code.

Part II | Home Assignments**Essay Questions:**

- What is the role of status code 500 in communication between the client and the server?
.....
.....
- What is an API, and how does it help in developing web applications?
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.....

Multiple-Choice Questions: (choose the correct answer)

1. An API is:
 - A. A database.
 - B. A set of rules for communication between programs.
 - C. A security protocol.
 - D. A data format.
2. When displaying search results from the server, what is usually used?
 - A. POST
 - B. GET
 - C. 500
 - D. HTTPS

PERIOD 3**Weekly Assessments**

Week:

Date:/...../.....

MODEL A**Essay Questions:**

1. Explain the client-server model.

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2. Why must HTTPS be used on a login page?

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3. What is the role of an API in web applications?

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4. Explain status code 200.

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Multiple-Choice Questions: (choose the correct answer)

1. Which of the following represents the role of the client in a web application?
 - A. Processing requests on the server.
 - B. Sending requests to get data.
 - C. Storing data in the database.
 - D. Determining the target audience for posts.
2. Which of the following is usually used to request a resource or data from the server?
 - A. POST
 - B. GET
 - C. API
 - D. JSON
3. Which of the following is a common format for exchanging data between applications?
 - A. HTML
 - B. JSON
 - C. CSS
 - D. SQL

4. Status code 404 means:

- A. The request succeeded.
- B. An internal server error.
- C. The requested resource does not exist.
- D. The request was not sent.

MODEL B

Essay Questions:

1. Explain why the request-response model is fundamental in web applications, giving a practical example.

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2. Distinguish between using the GET method and the POST method, and identify suitable situations for each.

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3. Explain the importance of the HTTPS protocol in protecting data during transmission over the network.

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4. Design a suitable communication method for a task in a web application (such as registering a new user).

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Multiple-Choice Questions: (choose the correct answer)

1. Which of the following status codes indicates that the request succeeded?

- A. 200
- B. 404
- C. 500
- D. 302

2. Which of the following is usually used to send form data to the server?

- A. GET
- B. POST
- C. API
- D. JSON

3. Which interface allows applications to communicate and exchange data?

- A. HTML
- B. API
- C. CSS
- D. SQL

4. Choose the one that represents an example of a data format used to exchange information between applications.

- A. XML
- B. JSON
- C. PHP
- D. Java

MODEL C**Essay Questions:**

1. Explain how the server handles requests coming from the client, and what happens if the request is invalid or not found.
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2. Explain the role of application programming interfaces (APIs) in connecting different applications, with a practical example.
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3. Explain how the JSON format is used in exchanging data, and why it is considered suitable for web applications.
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4. Suggest a scenario for a web application that needs to use the POST method, and explain the reason for your choice.
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Multiple-Choice Questions: (choose the correct answer)

1. Which of the following status codes indicates an internal server error?
A. 200
B. 404
C. 500
D. 302
2. Which of these technologies is usually used to request a web page or search results?
A. POST
B. GET
C. API
D. JSON
3. What text format is used to represent structured data using objects and arrays?
A. HTML
B. JSON
C. CSS
D. SQL
4. Which of the following is used to secure communication between the client and the server and prevent eavesdropping?
A. HTTP
B. HTTPS
C. API
D. JSON

UNIT 3 | WEB APPLICATIONS

LESSON

3

Fundamentals of Front-End Technology

PERIOD 1

Week:

Date:/...../.....

Part I | Performance Tasks

Essay Questions:

1. Explain the role of each of HTML, CSS, and JavaScript in building a web page.

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2. Why is semantic HTML important in building web pages?

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Multiple-Choice Questions: (choose the correct answer)

1. HTML is responsible for:

- | | |
|------------------------------|-------------------------------------|
| A. The visual appearance. | B. The basic structure of the page. |
| C. The interactive behavior. | D. Improving performance. |

2. Which of the following is a semantic HTML element, except...

- | | |
|-------------|-------------|
| A. <header> | B. <E-mail> |
| C. <main> | D. <footer> |

Part II | Home Assignments

Essay Questions:

1. Explain how responsive design helps improve the user experience.

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2. Explain the benefit of using a framework in front-end development.

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Multiple-Choice Questions: (choose the correct answer)

1. CSS is used for:
 - A. Building the structure.
 - B. Adding interactivity.
 - C. Formatting the appearance.
 - D. Managing databases.
2. JavaScript adds:
 - A. Headings and paragraphs.
 - B. Colors and fonts.
 - C. Interactive behavior.
 - D. Semantic elements.

PERIOD 2

Week:

Date:/...../.....

Part I | Performance Tasks**Essay Questions:**

1. Give an example of a semantic HTML element and write its function.

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2. Discuss how CSS contributes to improving the visual appearance of a page.

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Multiple-Choice Questions: (choose the correct answer)

1. The <header> element in semantic HTML is used for:
 - A. The main content.
 - B. Navigation.
 - C. The page header.
 - D. The bottom of the page.
2. Responsive design means:
 - A. Adjusting the layout according to different screen sizes.
 - B. Using JavaScript only.
 - C. Improving databases.
 - D. Hiding unimportant content.

Part II | Home Assignments**Essay Questions:**

1. What is the difference between the React library and the Next.js framework?

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2. Why is the mobile-first approach important in responsive design?

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Multiple-Choice Questions: (choose the correct answer)

- The React framework is used for:
 - Building user interfaces with reusable components.
 - Managing databases.
 - Encrypting the connection.
 - Improving performance only.
- The Next.js framework is built on:
 - Vue
 - React
 - CSS
 - HTML

PERIOD 3**Weekly Assessments**

Week:

Date:/...../.....

MODEL A**Essay Questions:**

- Discuss the role of HTML in building a web page.

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- Explain how CSS helps improve the appearance.

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- What is responsive design?

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- What is the benefit of using a framework?

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Multiple-Choice Questions: (choose the correct answer)

- Used to determine the appearance and formatting of a web page?
 - HTML
 - CSS
 - JSON
 - JavaScript
- Which of the following HTML elements is used to define the main section of the content?
 - <header>
 - <main>
 - <footer>
 - <nav>
- Responsive design means:
 - Making the page work only on computers.
 - Automatically adjusting the layout according to different screen sizes.
 - Using JavaScript to add interactivity.
 - Improving server speed.

4. Which of the following is a JavaScript library for building user interfaces?

- A. React
- C. CSS

- B. HTML
- D. SQL

MODEL B

Essay Questions:

1. Explain how HTML, CSS, and JavaScript work together to form a complete web page.

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2. Give an example of a semantic HTML element, and explain its benefit in improving accessibility.

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3. Explain the concept of “mobile-first” in responsive design, and why it is used.

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4. Name two common libraries and frameworks for web development, explaining their characteristics.

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Multiple-Choice Questions: (choose the correct answer)

1. Which of the following technologies is used to build the basic structure of the page?

- A. CSS
- C. JavaScript

- B. HTML
- D. SQL

2. The <nav> element in HTML is used for:

- A. Displaying the main content.
- C. Displaying the main heading.

- B. Navigating between parts of the site.
- D. Displaying the footer.

3. The basic advantage of responsive design is:

- A. Improving server speed.
- C. Adding interactivity using JavaScript.

- B. Adapting to different screen sizes.
- D. Improving search engines.

4. A framework built on React that provides routing and page-rendering methods is:

- A. Next.js
- C. CSS

- B. Vue
- D. HTML

MODEL C

Essay Questions:

1. How does JavaScript add interactivity to a web page? Give a practical example.

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2. State the benefits of using semantic HTML for search engines.

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3. Explain how responsive design helps improve the user experience on smartphones.

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4. Discuss the role of the Next.js framework in developing modern web applications.

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Multiple-Choice Questions: (choose the correct answer)

1. Which of the following technologies is used to add interactive behavior to the page?

A. HTML
B. CSS
C. JavaScript
D. SQL

2. The <footer> element in HTML is used for:

A. Displaying the main content.
B. Displaying the footer at the bottom of the page.
C. Displaying the main heading.
D. Navigating between parts of the site.

3. The term “mobile-first” in responsive design means:

A. Designing the page first for computers.
B. Designing the page first for small screens such as phones.
C. Using JavaScript to add interactivity.
D. Improving server speed.

4. Which of the following is a JavaScript framework that allows applications to be built incrementally?

A. React
B. Next.js
C. Vue
D. HTML